



ECKERLE

# Extented 8D-Report

Doc No.:RD-QP-24-3 V0.1

| General Information           |                                                                                        |
|-------------------------------|----------------------------------------------------------------------------------------|
| Customer                      | Eckerle (Zhuhai) Co., Ltd.                                                             |
| Contact customer              | Li Zhefeng                                                                             |
| Tel. No.                      | 0756-6808667                                                                           |
| Email address                 | <a href="mailto:Zhefeng.Li@cn.eckerle-gruppe.com">Zhefeng.Li@cn.eckerle-gruppe.com</a> |
| Complaint_No.:                | 20260401                                                                               |
| Detection Date:               | 2026-04-01                                                                             |
| Part Number:                  | 3134320725                                                                             |
| Designation:                  | carbon brush                                                                           |
| Issue date of the 8D-Report:  |                                                                                        |
| Update date of the 8D-Report: |                                                                                        |
| Final date of the 8D-Report:  |                                                                                        |

In

| D1 Problem solution team(Mark person in charge with *) |           |        |                 |
|--------------------------------------------------------|-----------|--------|-----------------|
| Supplier: AVO Korea                                    |           |        |                 |
| Name:                                                  | Position: | Phone: | E-mail Adresse: |
|                                                        |           |        |                 |

| D2 Description of problem                                                                                                                                                                                                                                                                                                                                                                 |                         |         |               |                  |                     |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|---------|---------------|------------------|---------------------|
| During our assembly process we found that unknown carbon brushes were mixed in the GBM carbon brush 3134320725, please confirm the failure and submit the analysis report to us. Thank you.                                                                                                                                                                                               |                         |         |               |                  |                     |
| The basic information as following.                                                                                                                                                                                                                                                                                                                                                       |                         |         |               |                  |                     |
| <b>Date:</b> 2026-04-01                                                                                                                                                                                                                                                                                                                                                                   |                         |         |               |                  |                     |
| <b>Location:</b> GBM Line                                                                                                                                                                                                                                                                                                                                                                 |                         |         |               |                  |                     |
| <b>Who:</b> Inspector                                                                                                                                                                                                                                                                                                                                                                     |                         |         |               |                  |                     |
| <b>PN:</b> 3134320725                                                                                                                                                                                                                                                                                                                                                                     |                         |         |               |                  |                     |
| <b>Name:</b> GBM Carbon brush                                                                                                                                                                                                                                                                                                                                                             |                         |         |               |                  |                     |
| <b>Failure mode/QTY:</b> unknown carbon brushes mixed 26 pcs                                                                                                                                                                                                                                                                                                                              |                         |         |               |                  |                     |
| <b>Blocked QTY:</b>                                                                                                                                                                                                                                                                                                                                                                       |                         |         |               |                  |                     |
| PN                                                                                                                                                                                                                                                                                                                                                                                        | Batch                   | AVO LOT | Incoming qty  | Blocked qty      |                     |
|                                                                                                                                                                                                                                                                                                                                                                                           |                         |         |               | finished product | parts               |
| 3134320725                                                                                                                                                                                                                                                                                                                                                                                | B260309015              | K26B24  | 10000         | -                | -                   |
| Delivery information:                                                                                                                                                                                                                                                                                                                                                                     |                         |         |               |                  |                     |
| date                                                                                                                                                                                                                                                                                                                                                                                      | Incoming document no    | PO no.  | PN            | ECKERLE lot      | Quantity in invoice |
| 2026/3/9                                                                                                                                                                                                                                                                                                                                                                                  | AV026057 IP202603055UPS | -       | 3 134 320 725 | B260309015       | 10,000              |
| Requirement from us:                                                                                                                                                                                                                                                                                                                                                                      |                         |         |               |                  |                     |
| 1, Confirm failure, if you need samples let me know.                                                                                                                                                                                                                                                                                                                                      |                         |         |               |                  |                     |
| 2, Take containment action(s) immediately in your plant;                                                                                                                                                                                                                                                                                                                                  |                         |         |               |                  |                     |
| 3, Advise us how to handle the affected parts in whole supply chain within two working days, Overdue we will seem you don't follow BOSCH quality agreement and 8D rules , and we have to initiate the right authorized by Bosch to take effective containment actions for your failure and charge you all the possible cost to keep customer line running, and automatically it seems you |                         |         |               |                  |                     |



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Doc No.:RD-QP-24-3 V0.1

have already given up the right to make any arguments for the failure and cost at moment or in future. Furthermore, we have to escalate to Bosch management level to take next step actions in business level. Pls kindly note that all the relating cost will charge from your payment.

4, improvement actions must be taken accordingly.

5, Submit 8D report and strictly follow rules as: D3 within 2 days. D5 within 10days, D6 within 30days and D8 within 40days.

6, Offer for sorting costs.

|                   |            |  |
|-------------------|------------|--|
| Part No.          | 3134320725 |  |
| Batch No          |            |  |
| status            |            |  |
| Sorting qty       |            |  |
| Efficiency        |            |  |
| sorting time      |            |  |
| hourly cost       |            |  |
| Sorting cost      |            |  |
| Failure qty       |            |  |
| Price(CNY/100pcs) |            |  |
| Scrap cost (EUR)  |            |  |
| TOTAL COST /EUR   |            |  |

- The unit price includes Import duties and Import transport cost.
- FX(USD/RMB)= 7.24

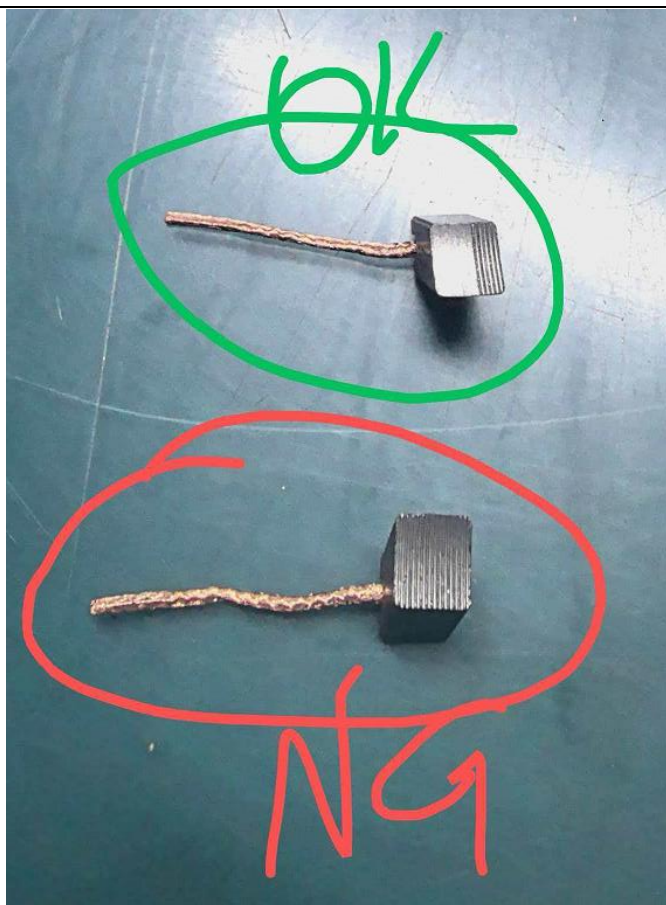
**Failure picture**



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# Extended 8D-Report

Doc No.:RD-QP-24-3 V0.1



Label



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# Extended 8D-Report

Doc No.:RD-QP-24-3 V0.1

|          |                |           |              |
|----------|----------------|-----------|--------------|
| Customer | Eckerle Zhuhai | Part Name | Carbon Brush |
| Part No. | 3134320725     | Lot No.   | K26B24       |
| Quantity | 5,000EA        | Date      | 2026.02.24   |

## D3 Short term action

| Action: | Responsible | Date | Status |
|---------|-------------|------|--------|
| 1.      |             |      |        |
| 2.      |             |      |        |
| 3.      |             |      |        |
| 4.      |             |      |        |
| 5.      |             |      |        |
| 6.      |             |      |        |

## Risk estimate

T.

## Process description

| Process step | Description | See Ishikawa |
|--------------|-------------|--------------|
|--------------|-------------|--------------|

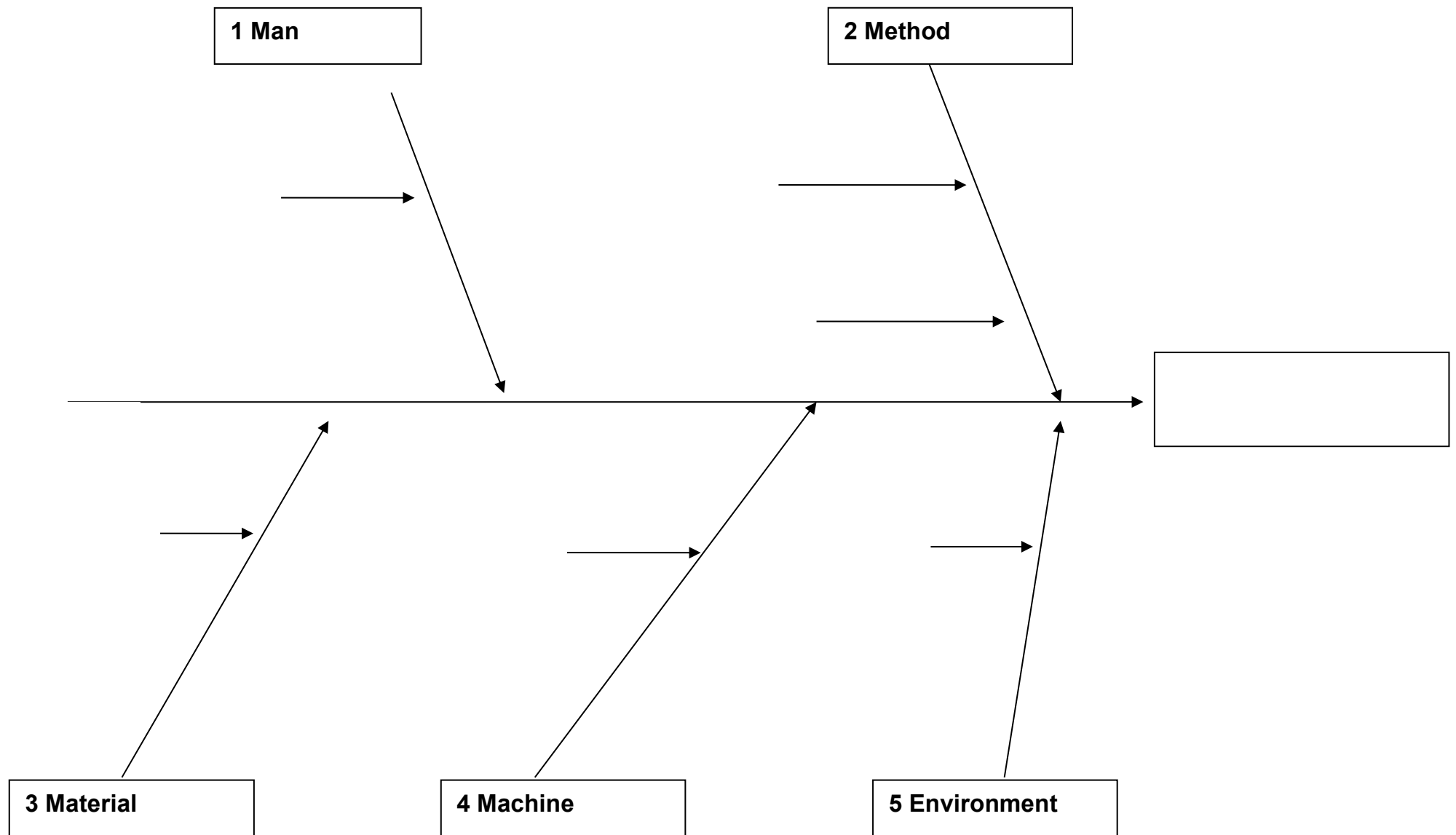


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Doc No.:RD-QP-24-3 V0.1

|  |  |  |
|--|--|--|
|  |  |  |
|  |  |  |
|  |  |  |
|  |  |  |
|  |  |  |
|  |  |  |
|  |  |  |





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Doc No.:RD-QP-24-3 V0.1

| 5 M            | Possible root cause | Does not fit, because: | Analysis by 5W |
|----------------|---------------------|------------------------|----------------|
| 1. Man         |                     |                        |                |
|                |                     |                        |                |
| 2. Method      |                     |                        |                |
|                |                     |                        |                |
| 3. Material    |                     |                        |                |
|                |                     |                        |                |
| 4. Machine     |                     |                        |                |
|                |                     |                        |                |
|                |                     |                        |                |
|                |                     |                        |                |
| 5. Environment |                     |                        |                |
|                | 5.2                 |                        |                |



# Extended 8D-Report

Doc No.: RD-QP-24-3 V0.1

|                                        |  |
|----------------------------------------|--|
| <b>5x-Why-Analysis to 4.1.3</b>        |  |
| <b>Why has this failure occurred?</b>  |  |
| Why ?                                  |  |
| Why ?                                  |  |
| Why ?                                  |  |
| Why ?                                  |  |
| Why ?                                  |  |
| <b>Why was the fault not detected?</b> |  |
| Why ?                                  |  |
| Why ?                                  |  |
| Why ?                                  |  |
| Why ?                                  |  |
| Why ?                                  |  |

|                            |                    |             |               |
|----------------------------|--------------------|-------------|---------------|
| <b>D5 Possible actions</b> |                    |             |               |
| <b>Process Step:</b>       |                    |             |               |
| <b>Fault prevention</b>    | <b>Responsible</b> | <b>Date</b> | <b>Status</b> |
|                            |                    |             |               |
| <b>Fault detection</b>     | <b>Responsible</b> | <b>Date</b> | <b>Status</b> |
|                            |                    |             |               |
| <b>Fault prevention</b>    | <b>Responsible</b> | <b>Date</b> | <b>Status</b> |
|                            |                    |             |               |
| <b>Fault detection</b>     | <b>Responsible</b> | <b>Date</b> | <b>Status</b> |
|                            |                    |             |               |

|                                      |                    |             |               |
|--------------------------------------|--------------------|-------------|---------------|
| <b>D6 Long term action</b>           |                    |             |               |
| <b>Process Step:</b>                 |                    |             |               |
| <b>Fault prevention</b>              | <b>Responsible</b> | <b>Date</b> | <b>Status</b> |
|                                      |                    |             |               |
| <b>Verification of effectiveness</b> | <b>Responsible</b> | <b>Date</b> | <b>Status</b> |
|                                      |                    |             |               |
| <b>Fault detection</b>               | <b>Responsible</b> | <b>Date</b> | <b>Status</b> |
|                                      |                    |             |               |





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Doc No.:RD-QP-24-3 V0.1

|                                      |                    |             |               |
|--------------------------------------|--------------------|-------------|---------------|
|                                      |                    |             |               |
| <b>Verification of effectiveness</b> | <b>Responsible</b> | <b>Date</b> | <b>Status</b> |
|                                      |                    |             |               |

|                 |                    |                    |             |               |
|-----------------|--------------------|--------------------|-------------|---------------|
| <b>Document</b> | <b>Description</b> | <b>Responsible</b> | <b>Date</b> | <b>Status</b> |
|                 |                    |                    |             |               |
|                 |                    |                    |             |               |
|                 |                    |                    |             |               |
|                 |                    |                    |             |               |

**Pictures situation before and after implementation of actions**

**Before:**

**After:**

**D7 Preventive action**

**Transfer of actions (productgroups, part numbers, ....)**



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Doc No.:RD-QP-24-3 V0.1

|                                      |                    |             |               |
|--------------------------------------|--------------------|-------------|---------------|
|                                      |                    |             |               |
| <b>Fault prevention</b>              | <b>Responsible</b> | <b>Date</b> | <b>Status</b> |
|                                      |                    |             |               |
| <b>Verification of effectiveness</b> | <b>Responsible</b> | <b>Date</b> | <b>Status</b> |
|                                      |                    |             |               |
| <b>Fault detection</b>               | <b>Responsible</b> | <b>Date</b> | <b>Status</b> |
|                                      |                    |             |               |
| <b>Verification of effectiveness</b> | <b>Responsible</b> | <b>Date</b> | <b>Status</b> |
|                                      |                    |             |               |

|                            |                  |              |                   |
|----------------------------|------------------|--------------|-------------------|
| <b>D8 Final discussion</b> |                  |              |                   |
| <b>Remarks</b>             |                  |              |                   |
|                            |                  |              |                   |
| <b>Name:</b>               | <b>Position:</b> | <b>Date:</b> | <b>Signature:</b> |
|                            |                  |              |                   |
|                            |                  |              |                   |
|                            |                  |              |                   |
|                            |                  |              |                   |